ÉCOLE SUPÉRIEURE FRANÇAISE
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COLLEGE DE PARIS

THEME

RAPPORT CONDUITE DE PROJET : DEVOPS, KANBAN, METHODE AGILE

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MASTER 1

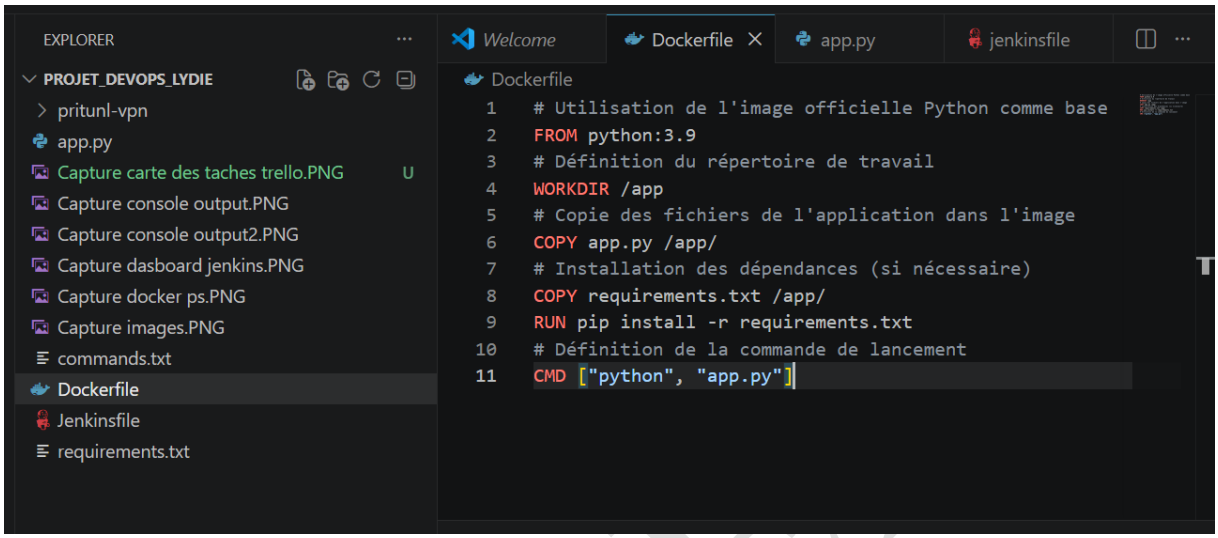
INTRODUCTION

Ce projet s'inscrit dans une démarche DevOps visant à automatiser le déploiement et la gestion d'une application web. Il consiste à développer une application Flask simple, puis à la conteneuriser à l'aide de Docker. Une chaîne d'intégration et de déploiement continu (CI/CD) est mise en place avec Jenkins afin d'automatiser les différentes étapes du processus. Par ailleurs, une infrastructure sécurisée est implémentée grâce à un serveur VPN Pritunl utilisant Docker Compose et MongoDB. Enfin, la gestion du projet est organisée à l'aide des outils Jira et Trello selon une approche agile Scrum.

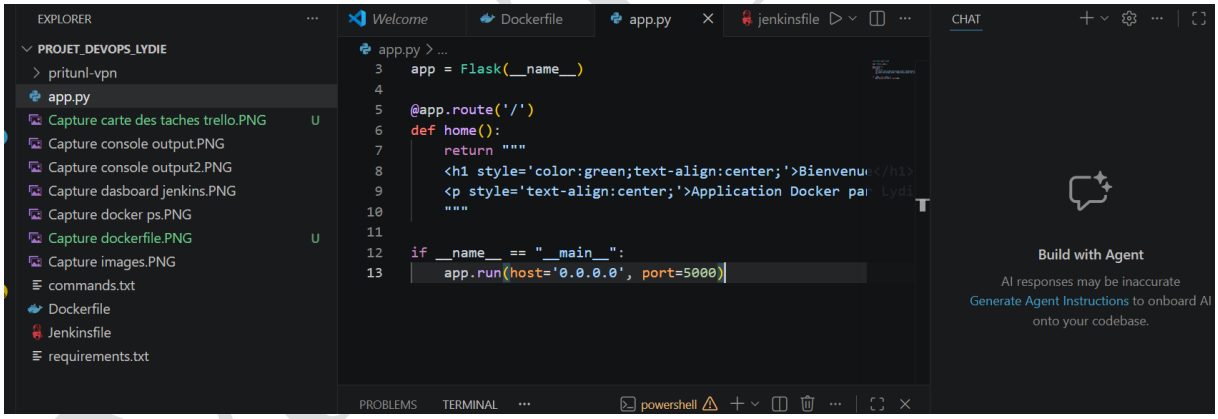
MASTER 1

I. TP1 DOCKERFILE

I.1. Création du fichier Docker file



I.2. Création du fichier app.py



I.3. Construction de l'image Docker hub

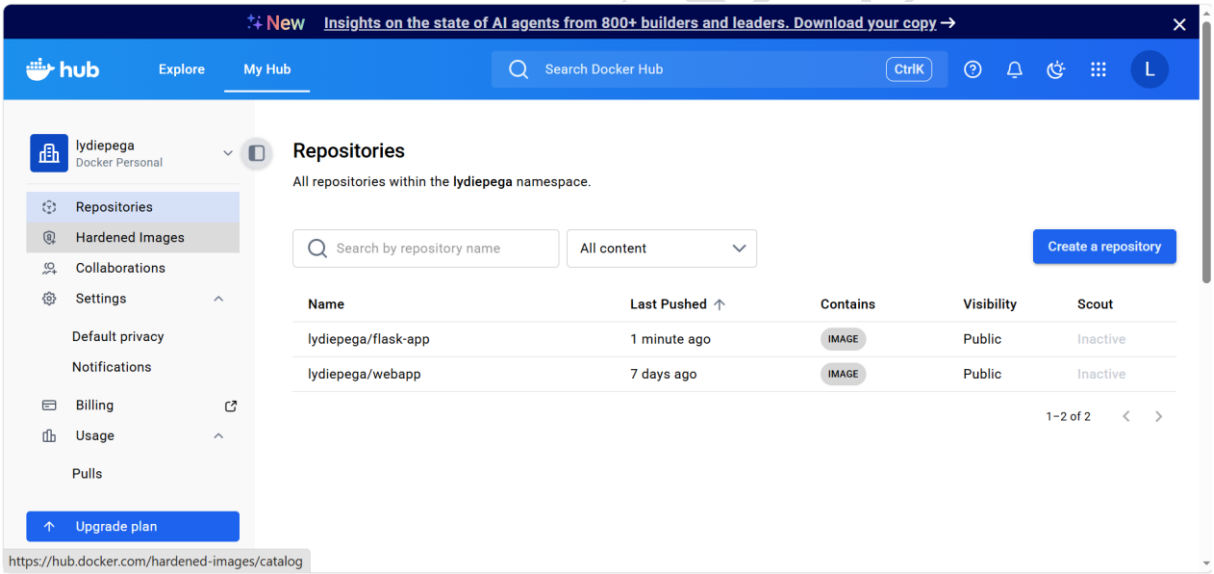
```
docker --version
docker build -t lydie-flask-app .
docker run -d -p 5000:5000 lydie-flask-app
docker ps
```

```
lydie@DESKTOP-LYDIEP:/mnt/e/cours M1- keyce/CONDUITE DE PROJET- AGILE, DEVOPS, KANBAN/Projet_DevOps_Lydie$ sudo docker build -t flask-app .
 => [3/5] COPY app.py /app/                                                                                    0.1s
 => [4/5] COPY requirements.txt /app/                                                                          0.1s
 => [5/5] RUN pip install -r requirements.txt                                                                 15.6s
 => exporting to image                                                                                         2.3s
 => => exporting layers                                                                                        1.4s
 => => exporting manifest sha256:fe400d34699124d967166f0c3ca8ba2b945e7ec26e1eb37541636c2343566cb5              0.0s
 => => exporting config sha256:ea524eece1f48d5a0a277fca528af3a41b647f5a32880ba7a242e3f299b0bd72                0.0s
 => => exporting attestation manifest sha256:6fbedd158507d517d6eca4ba2a7aa44860ce91f9aaa71e880bc100c2dfcaeb7a  0.1s
 => => exporting manifest list sha256:ab61aef4f7684f08a5bd9f8ea357c26c4c6543928865980ed4fe2867fe8fb80e         0.0s
 => => naming to docker.io/library/flask-app:latest                                                            0.0s
 => => unpacking to docker.io/library/flask-app:latest                                                         0.6s
lydie@DESKTOP-LYDIEP:/mnt/e/cours M1- keyce/CONDUITE DE PROJET- AGILE, DEVOPS, KANBAN/Projet_DevOps_Lydie$
```

- **Vérification de l'image** :

```
lydie@DESKTOP-LYDIEP:/mnt/e/cours M1- keyce/CONDUITE DE PROJET- AGILE, DEVOPS, KANBAN/Projet_DevOps_Lydie$ sudo docker images
                                                                                                  Info →   U  In Use
IMAGE                  ID            DISK USAGE   CONTENT SIZE   EXTRA
flask-app:latest       ab61aef4f768       1.6GB          410MB
jippi/pritunl:latest   8898d9bf459d      2.62GB          670MB    U
mongo:6                03cda579c8ca      1.06GB          273MB    U
lydie@DESKTOP-LYDIEP:/mnt/e/cours M1- keyce/CONDUITE DE PROJET- AGILE, DEVOPS, KANBAN/Projet_DevOps_Lydie$
```

- Pousser l'image dans le docker hub en public :




```
sk-app:latest
lydie@DESKTOP-LYDIEP:/mnt/e/cours M1- keyce/CONDUITE DE PROJET- AGILE, DEVOPS, KANBAN/Projet_DevOps_Lydie$ sudo docker login
Authenticating with existing credentials... [Username: lydiepega]

 Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
lydie@DESKTOP-LYDIEP:/mnt/e/cours M1- keyce/CONDUITE DE PROJET- AGILE, DEVOPS, KANBAN/Projet_DevOps_Lydie$ sudo docker push lydiepega/flask-app:la
test
The push refers to repository [docker.io/lydiepega/flask-app]
c9723aa529b0: Pushed
dae148ed6d6b: Pushed
```

- Aperçu de l'application

```
lydie@DESKTOP-LYDIEP:/mnt/e/cours M1- keyce/CONDUITE DE PROJET- AGILE, DEVOPS, KANBAN/Projet_DevOps_Lydie$ sudo docker images
[sudo] password for lydie:
                                                                                                    i Info →   U In Use
IMAGE                          ID             DISK USAGE   CONTENT SIZE   EXTRA
flask-app:latest               ab61aef4f768       1.6GB          410MB
jippi/pritunl:latest           8898d9bf459d      2.62GB          670MB     U
lydiepega/flask-app:latest     ab61aef4f768       1.6GB          410MB
mongo:6                        03cda579c8ca      1.06GB          273MB     U
lydie@DESKTOP-LYDIEP:/mnt/e/cours M1- keyce/CONDUITE DE PROJET- AGILE, DEVOPS, KANBAN/Projet_DevOps_Lydie$
```

Lien GitHub : lydie-dev/Projet_DevOps_Lydie

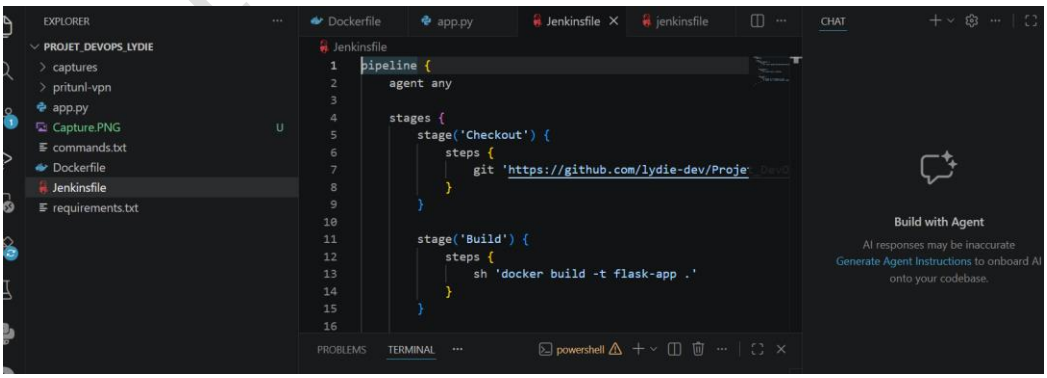
Lien de l'image sur Docker Hub : Repositories | lydiepega

II. TP2 : CI/CD

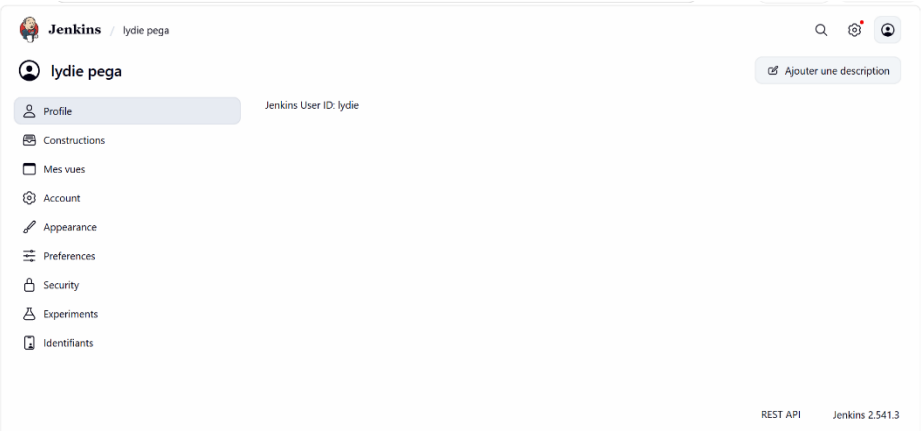
I.1. Création d'un fichier Jenkins

```
lydie@DESKTOP-LYDIEP:~$ sudo docker exec -u root -it jenkins bash
Processing triggers for libc-bin (2.41-12) ...
Processing triggers for systemd (257.9-1~deb13u1) ...
root@f91232f4ed8f:/# docker --version
Docker version 26.1.5+dfsg1, build a72d7cd
root@f91232f4ed8f:/# docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED         STATUS          PORTS
                    NAMES
f91232f4ed8f   jenkins/jenkins:lts    "/usr/bin/tini -- /u…"   14 minutes ago  Up 14 minutes   0.0.0.0:8080->8080/tcp, :::8080->8080/tcp, 0.0.0.0:50000->50000/tcp,
 :::50000->50000/tcp   jenkins
e487dc7f4d4a   lydie-flask-app        "python app.py"          2 hours ago     Up 2 hours      0.0.0.0:5000->5000/tcp, :::5000->5000/tcp
                    crazy_sanderson
root@f91232f4ed8f:/#
```

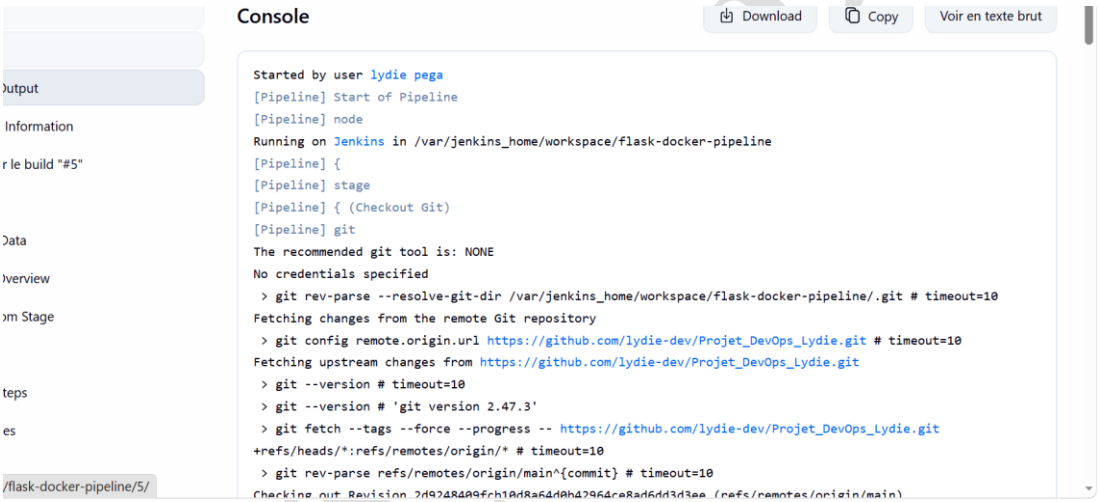
- fichier pipeline.yml



➢ Dashboard Jenkins



➢ Console Output



```
lydie@DESKTOP-LYDIEP:~$ sudo docker ps
                                    sad_maxwell
f91232f4ed8f   jenkins/jenkins:lts   "/usr/bin/tini -- /u…"   42 minutes ago   Up 11 minutes   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tc
p, [::]:50000->50000/tcp   jenkins
e487dc7f4d4a   lydie-flask-app       "python app.py"          2 hours ago      Up 2 hours      0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
                                    crazy_sanderson
lydie@DESKTOP-LYDIEP:~$ sudo docker images
                                                                                                  Info →   U  In Use
IMAGE                    ID             DISK USAGE   CONTENT SIZE   EXTRA
flask-app:latest         cc6d0d4b11f0       1.6GB          410MB     U
jenkins/jenkins:lts      c4098086090c       806MB          288MB     U
lydie-flask-app:latest   8873e06bcacf       1.6GB          410MB     U
lydie@DESKTOP-LYDIEP:~$
```

III. Docker compose

```
lydie@DESKTOP-LYDIEP:/mnt/e/cours M1- keyce/CONDUITE DE PROJET- AGILE, DEVOPS, KANBAN/Projet_DevOps_Lydie$ sudo docker pull mongo:6
sudo docker pull jippi/pritunl
Status: Downloaded newer image for jippi/pritunl:latest
docker.io/jippi/pritunl:latest
lydie@DESKTOP-LYDIEP:/mnt/e/cours M1- keyce/CONDUITE DE PROJET- AGILE, DEVOPS, KANBAN/Projet_DevOps_Lydie$ cd pritunl-vpn
sudo docker compose up -d
[+] up 5/5
 ✓ Network pritunl-vpn_vpn-net     Created                                                    0.3s
 ✓ Volume pritunl-vpn_mongo_data   Created                                                    0.0s
 ✓ Volume pritunl-vpn_pritunl_data Created                                                    0.0s
 ✓ Container mongo                 Started                                                    1.6s
 ✓ Container pritunl               Started                                                    2.9s
```

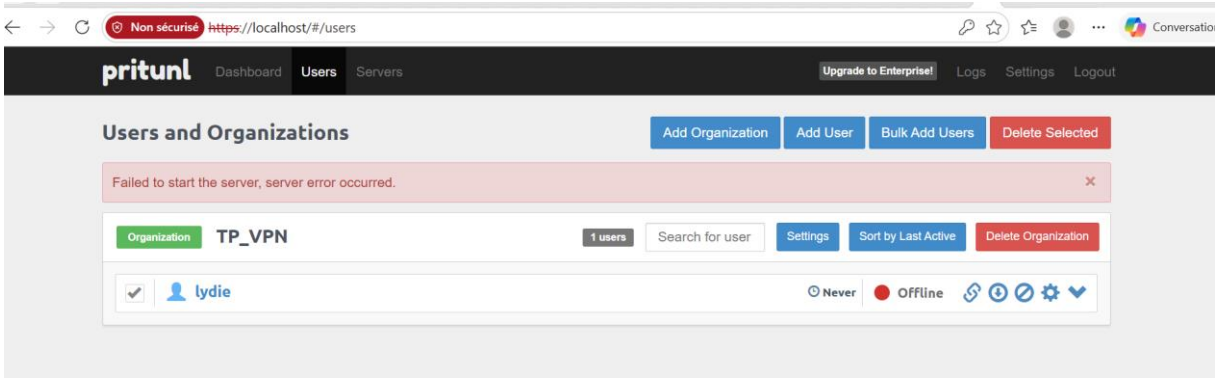
```
lydie@DESKTOP-LYDIEP:/mnt/e/cours M1- keyce/CONDUITE DE PROJET- AGILE, DEVOPS, KANBAN/Projet_DevOps_Lydie/pritunl-vpn$ sudo docker exec -it pritunl bash
root@7f041b09f89c:/# pritunl default-password
[local][2026-04-15 18:41:29,059][INFO] Getting default administrator password
Administrator default password:
  username: "pritunl"
  password: "midS8E8uTiJM"
```

```
lydie@DESKTOP-LYDIEP:~$ sudo docker exec -u root -it jenkins bash
Processing triggers for libc-bin (2.41-12) ...
Processing triggers for systemd (257.9-1~deb13u1) ...
root@f91232f4ed8f:/# docker --version
Docker version 26.1.5+dfsg1, build a72d7cd
root@f91232f4ed8f:/# docker ps
CONTAINER ID   IMAGE                 COMMAND                  CREATED          STATUS          PORTS
                        NAMES
f91232f4ed8f   jenkins/jenkins:lts   "/usr/bin/tini -- /u…"   14 minutes ago   Up 14 minutes   0.0.0.0:8080->8080/tcp, :::8080->8080/tcp, 0.0.0.0:50000->50000/tcp,
 :::50000->50000/tcp   jenkins
e487dc7f4d4a   lydie-flask-app       "python app.py"          2 hours ago      Up 2 hours      0.0.0.0:5000->5000/tcp, :::5000->5000/tcp
                        crazy_sanderson
root@f91232f4ed8f:/#
```

- **Test du vpn avec docker compose :**

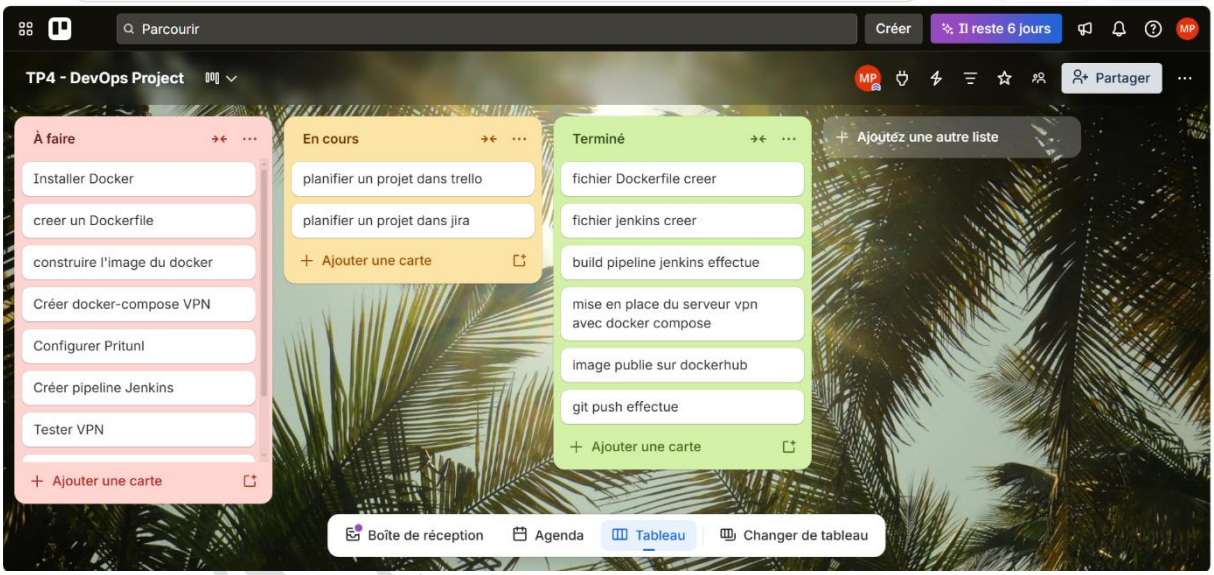
```
lydie@DESKTOP-LYDIEP:/mnt/e/cours M1- keyce/CONDUITE DE PROJET- AGILE, DEVOPS, KANBAN/Projet_DevOps_Lydie/pritunl-vpn$ ls
'Capture printl vpn.PNG'   TP_VPN_lydie_vpn-server.ovpn   data   docker-compose.yml   mongodb
lydie@DESKTOP-LYDIEP:/mnt/e/cours M1- keyce/CONDUITE DE PROJET- AGILE, DEVOPS, KANBAN/Projet_DevOps_Lydie/pritunl-vpn$ sudo openvpn --config TP_VPN_lydi
e_vpn-server.ovpn
2026-04-16 02:09:01 us=112686 Note: cipher 'AES-128-CBC' in --data-ciphers is not supported by ovpn-dco, disabling data channel offload.
2026-04-16 02:09:01 us=112903 Current Parameter Settings:
2026-04-16 02:09:01 us=112922 NOTE: --mute triggered...
2026-04-16 02:09:01 us=112944 291 variation(s) on previous 4 message(s) suppressed by --mute
2026-04-16 02:09:01 us=113004 OpenVPN 2.6.19 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [LZ4] [EPOLL] [PKCS11] [MH/PKTINFO] [AEAD] [DCO]
2026-04-16 02:09:01 us=113095 library versions: OpenSSL 3.0.13 30 Jan 2024, LZO 2.10
2026-04-16 02:09:01 us=113197 DCO version: N/A
```

- **Interface de printunl :**



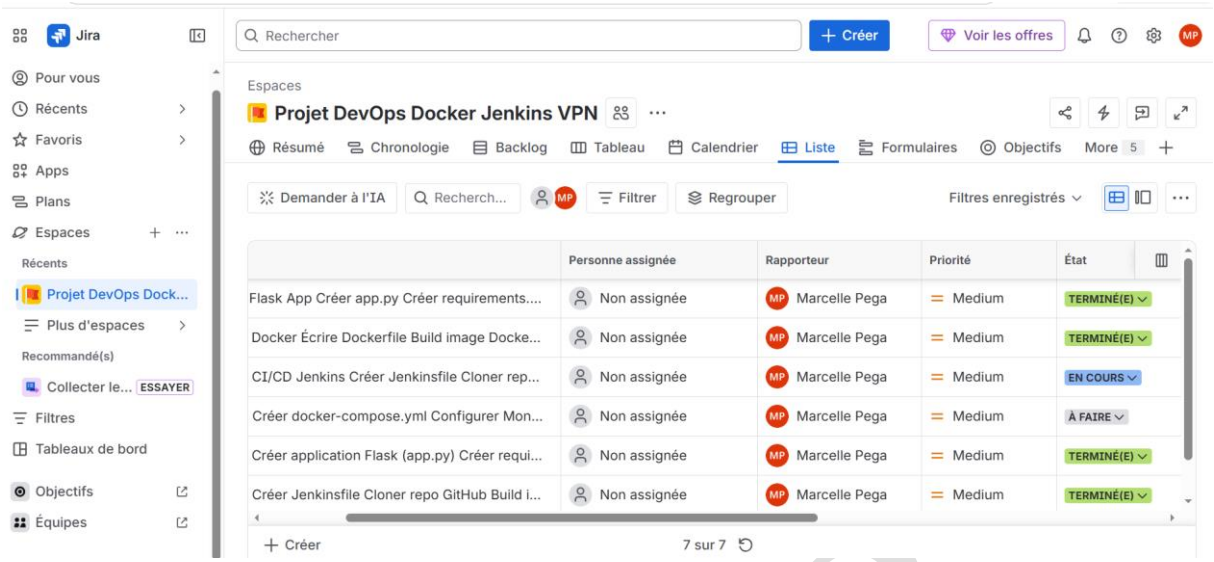
IV. Planification des projets dans Trello et jira

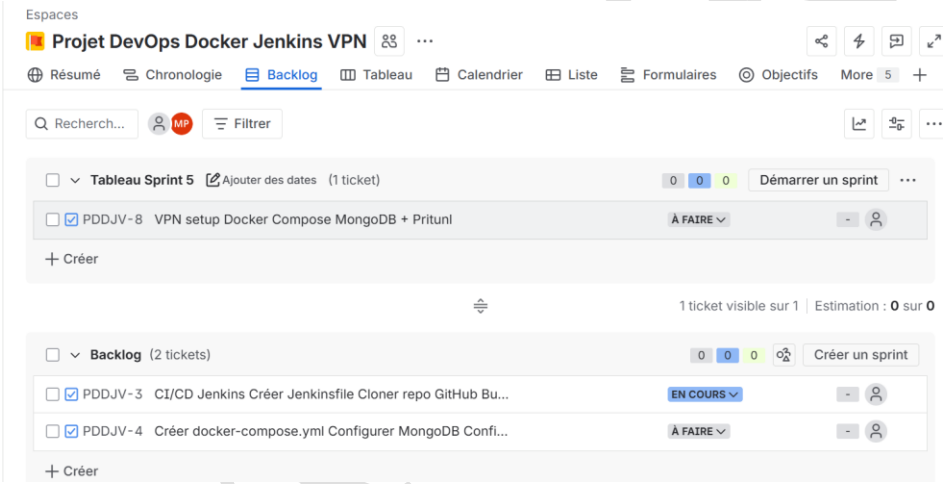
- **Dans Trello :**



Lien projet Trello : [TP4 - DevOps Project | Trello]

- **Dans Jira :**





Lien Jira : [Projet DevOps Docker Jenkins VPN – Backlog - Jira]

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